EXPERIMENT 6

DEMONSTRATING CHEMICAL EQUILIBRIUM USING LE CHÂTELIER'S PRINCIPLE

INTRODUCTION

When we begin to study chemical reactions, we often assume that the reaction goes to completion. We assume that when reactants are added together, the reaction continues from left to right until one of the reactants (the limiting reagent) is completely used up and the reaction stops. An example is the neutralization reaction that occurs when solutions of hydrochloric acid and sodium hydroxide are added together:

$HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$ [Rxn 1a]

In Reaction 1a, hydrochloric acid reacts with sodium hydroxide to form a salt (NaCl) and a molecule of water. Because hydrochloric acid is a strong acid and sodium hydroxide is a strong base, they react completely to neutralize each other and there is essentially no tendency for the reverse reaction to occur. Looking at it another way, when sodium chloride is dissolved in water (the reverse of Reaction 1a), there is no reaction. That is, hydrochloric acid and sodium hydroxide do not form in this solution.

$NaCl_{(aq)} + H_2O_{(l)} \not\rightarrow HCl_{(aq)} + NaOH_{(aq)}$ [Rxn 1b]

Notice that the arrow used in Reaction 1a points in one direction – left to right – to indicate that this reaction essentially goes to completion in the direction in which it is written.

Often, however, reactions stop before reaching completion. There is a forward reaction by which products are formed, and there is a reverse reaction by which reactants are re-formed. The two reactions take place simultaneously until the system reaches **equilibrium.** At this point, both the forward and reverse reactions are taking place at the same rate, so the amounts of reactants and products stay constant even though the chemicals species are continuing to react and re-form. We describe the reaction as being in a state of **dynamic equilibrium.** An example of a reaction at equilibrium is the reaction between solutions of iron (III) nitrate and potassium thiocyanate:

$Fe(\mathbf{NO_3})_{3(aq)} + \mathbf{K}SCN_{(aq)} \leftrightarrows [FeNCS]^{2+}_{(aq)} + 3\mathbf{KNO_3}_{(aq)}$ [Rxn 2]

Since nitrate and potassium ions (in bold above) are spectators, we can simplify the reaction by considering the net ionic equation:

$Fe^{3+}_{(aq)} + SCN^-_{(aq)} \leftrightarrows [FeNCS]^{2+}_{(aq)}$ [Rxn 3]

Notice that in Reaction 3, the thiocyanate ion "flips" end for end when it bonds to Fe^{3+} ions so that the nitrogen end of the SCN^- ion bonds to the metal ion. In addition, the product of the reaction has a bracket around it, which indicates that it is a **complex ion**. A complex ion is a cation that forms from the reaction of a metal ion (e.g., Fe^{3+} or Co^{2+}) with a nonmetal species (e.g., SCN^- or Cl^-). The proper chemical name for the product in Reaction 3 is isothiocyanatoiron (III).

You should also notice that a double arrow (⇆) is used in Reaction 3 to indicate that both the forward and reverse reactions are occurring simultaneously.

How do we know that both reactions are occurring at the same time? Concentrated solutions of $Fe^{3+}_{(aq)}$ are light yellow and concentrated solutions containing $[FeNCS]^{2+}_{(aq)}$ ions are an intense orange/red color. At lower concentration, $Fe^{3+}_{(aq)}$ solutions sometimes appear to be almost colorless and those containing $[FeNCS]^{2+}_{(aq)}$ ions are orange/red. Thiocyanate solutions are colorless.

When the nearly colorless dilute $Fe^{3+}_{(aq)}$ solution is added to colorless $SCN^{-}_{(aq)}$, an intense orange/red solution results. If the color of the solution stabilizes and does not become more intense or less intense (darker or lighter) over time, one can assume that the solution has reached equilibrium.

DISTURBING THE SYSTEM AT EQUILIBRIUM

Le Châtelier's Principle states that when a system at equilibrium is "stressed" by a disturbance, such as changing the concentration of a reactant or a product, or changing the temperature of the solution, the system will adjust in order to relieve the stress and establish a new position of equilibrium.

In this experiment, we will study two reactions at equilibrium. We will disturb the equilibrium in one of two ways. In both Parts A and B, the concentrations of reactants and/or products will be changed. In addition, the solution in Part B will be heated and cooled to observe the effects of stressing the equilibrium by changing the temperature. The color changes that occur because of these disturbances will allow us to determine the direction in which the equilibrium shifts in order to minimize the stress on the system.

It should be noted that although the volumes of solutions will increase when various reagents are added together, these effects are not as significant as the impact of adding reagents at high concentrations. You should neglect volume effects and focus on the changes in moles of reactants and/or products added to disturb the equilibrium of the system.

PART A: $Fe^{3+}_{(aq)} + SCN^{-}_{(aq)} \leftrightarrows [FeNCS]^{2+}_{(aq)}$

In the first part of the experiment, a concentrated solution of $Fe^{3+}_{(aq)}$ will be added to a concentrated solution of $SCN^{-}_{(aq)}$ in a wellplate. The $[FeNCS]^{2+}$ complex ion will form immediately and the solution will turn deep orange/brown. Then this solution will be diluted so that the color is not as intense. Various disturbances or stresses on the system at equilibrium will be carried out in several different wells.

Looking at Reaction 3, one might predict that if more $Fe^{3+}_{(aq)}$ or $SCN^{-}_{(aq)}$ ions were added, more orange-colored $[FeNCS]^{2+}_{(aq)}$ ions would be formed and the difference in intensity would be easily discernible. Conversely, if either of the reactants were removed from the solution, one might predict that the orange/red color of the $[FeNCS]^{2+}$ solution would become fainter or might disappear altogether.

PART B: Pink Reactant ⇆ Blue Product

For simplicity, the chemicals in this second equilibrium system will be described only by their colors: a pink reactant, a blue product, and a mixture of reactant and product that is a purplish color. The reaction is represented by ONE of the following chemical equations where you must determine whether the reaction is exothermic or endothermic.

Pink Reactant ⇆ Blue Product + heat **[Rxn 4a]**

Pink Reactant + heat ⇆ Blue Product **[Rxn 4b]**

In part A you considered adding or removing reactants or products to possibly cause a disturbance to the system at equilibrium. The system would then adjust to relieve the stress. Now in part B you consider the energetic component: heat energy. Reaction 4a shows that heat is <u>generated</u> in an **exothermic** reaction. <u>Generation</u> of heat raises the temperature. Reaction 4b shows that heat is *removed* in an **endothermic** reaction. *Removal* of heat causes cooling. Similar to the stress put on a chemical system by changing concentrations of species, a change in temperature of the system at equilibrium (by adding more heat or by cooling) would likewise disturb it. One could predict the direction in which the reaction is shifted, and the color changes that were observed would confirm the prediction.

Note on word choice: We suggest <u>generation</u> and *removal* of heat rather than release and absorption. The latter suggest movement of heat, but instead there is actual conversion of energy between heat (kinetic energy) and potential energy.

PROCEDURE

In all of the transfer and mixing steps that follow, **it is very important to use the correct pipet and avoid contamination**. Each solution has its own pipet, including the deionized water used for dilution. Stock solutions have a test tube containing a labeled pipet taped to the bottle. Pipets for the solutions prepared in the wellplates are lined up in the test tube rack and labeled according to the well which contains the solution to be pipetted. Therefore, check the label on each pipet before placing solutions in wells, and return it to the appropriate stock bottle or lay it on a paper towel to prevent contact with the countertop or other pipets.

Before starting the experiment, look at the report sheet to see how the wells in the well plate will be filled in each part of the experiment.

SAFETY NOTES

1. Wear goggles at all times.
2. Concentrated hydrochloric acid (HCl) is a strong acid that is irritating to the skin. Avoid breathing the vapors. In case of contact, wash thoroughly with large amounts of water and inform your instructor.
3. The "Blue Product" in part B is a solution of $[CoCl_4]^{2-}{}_{(aq)}$ which has been prepared using concentrated hydrochloric acid. If any of the solutions containing Co^{2+} ions solutions are spilled on your skin or clothing, rinse with plenty of water.

PART A: $Fe^{3+}_{(aq)} + SCN^-_{(aq)} \leftrightarrows [FeNCS]^{2+}_{(aq)}$

1. Reacting Fe^{3+} with SCN^-

(1.1) Take the 1-mL graduated pipet labeled **Fe^{3+} STOCK** from the 0.1 M $Fe(NO_3)_{3(aq)}$ stock bottle and deliver 1.0 mL of solution into well A1. Note the color of the solution in the A1 box on the report sheet. Return the pipet to the stock bottle.

(1.2) Use the 1-mL graduated pipet labeled **SCN^- STOCK** to deliver 1.0 mL of 0.1 M $KSCN_{(aq)}$ into well D1. Note the color of this solution in the D1 box on the report sheet. Return the pipet to the stock bottle.

(1.3) From the test tube rack, take the **Fe^{3+} (A1)** labeled graduated pipet and transfer 5 drops of 0.1 M Fe^{3+} from well A1 into well B2.

(1.4) Then take the **SCN^- (D1)** labeled graduated pipet and transfer 5 drops of 0.1M $KSCN_{(aq)}$ from well D1 into well B2. **Be careful not to contaminate the pipets by touching the other solutions.** Note the color of the resulting solution in the B2 box on the report sheet.

2. Diluting $[FeNCS]^{2+}$

(2.1) Swirl the well plate gently to mix the solution in well B2. From the test tube rack, take the **CONC $[FeNCS]^{2+}$ (B2)** labeled pipet and transfer 2 drops of the dark solution from well B2 into well C2.

Put some deionized water in a 100-mL beaker. Using the **H_2O labeled** pipet, add 3 mL of deionized water in order to dilute the solution (fill it three times to the 1-mL mark). Note the change in the intensity of the color between wells B2 and C2. The diluted solution at equilibrium in well C2 will serve as the **control solution.**

(2.2) Stir the solution in well C2 and, using a pipet labeled **DIL $[FeNCS]^{2+}$ (C2)**, transfer 0.5 mL of this solution into each of the wells A3, B3, C3, and D3. Record the intensity and color of the solutions in wells A3–D3 in the boxes on the report sheet (they should be the same as the dilute control solution in well C2).

When disturbances are made to the system at equilibrium, changes in intensity (lighter/darker) should occur as the system adjusts. When compared to the intensity of the control solution, the direction of the shift in equilibrium can be determined. If a stress has been placed on the system at equilibrium and relieving the stress results in the production of more of the $[FeNCS]^{2+}$ ion, the solution in the test well will have a more intense orange/red color compared to the solution in well C2, and the reaction has shifted to the right. If the equilibrium shifts left, toward the formation of more of the reactants, the intensity of the solution in the test well will be less than that observed in the control solution in well C2.

3. Disturbing the equilibrium

Before adding the reagents below, predict the direction in which the reaction will shift when the system at equilibrium is disturbed. Make a prediction for each of the steps 3.1 through 3.4 before you begin the experiment and record your predictions on the report sheet.

(3.1) **Adding $Fe^{3+}_{(aq)}$. To well A3,** add 5 drops of $Fe^{3+}_{(aq)}$ from well A1 using the **Fe^{3+} (A1)** pipet from the test tube rack. Note on the report sheet what is observed after the addition. Compare the intensity of the color in the "control" well C2 and determine in which direction the equilibrium actually shifted to relieve the stress on the system.

(3.2) **Adding $SCN^-_{(aq)}$. To well B3,** add 5 drops of $SCN^-_{(aq)}$ from well D1, using the **SCN^- (D1)** pipet from the test tube rack. Note observations on the report sheet. Compare the intensity of the color in well B3 with the control solution in well C2 and determine the actual direction of the shift in equilibrium.

(3.3) **Add $AgNO_{3(aq)}$. To well C3,** add 10 drops of 0.1 M $AgNO_{3(aq)}$ from the stock bottle. In a previous experiment, you learned that when solutions of $Ag^+_{(aq)}$ and $SCN^-_{(aq)}$ are mixed, a white AgSCN precipitate forms. In the reaction we are studying, removing SCN^- by precipitating it with Ag^+ ions disturbs the equilibrium and causes the reaction to shift to relieve the stress on the system. Note observations on the report sheet, including whether or not cloudiness is observed and whether color changes or intensity changes have occurred.

(3.4) **Add solid $Fe(NO_3)_3$. To well D3,** use a scoopula to add a few granules of iron (III) nitrate nonahydrate, $Fe(NO_3)_3 \bullet 9H_2O$, and stir with a stirring rod to help the solid dissolve. Note the change in the intensity of the color of the solution compared to that of the control well C2.

e. When you have finished, rinse the well plate thoroughly in the sink with plenty of tap water until no traces of orange/red solutions can be detected in the wells. Then use the wash bottle with deionized water to rinse each cell two or three times to assure cleanliness.

PART B: Pink Reactant ⇆ Blue Product

The instructor will demonstrate the effect of changing temperature on the equilibrium position in this reaction. THREE samples of the equilibrium solution will be available:

- one cooled in an ice bath
- one at room temperature
- one heated in a warm water bath

Record the color of the equilibrium solution at each temperature.

Cleaning Up

1. Flush all solutions down the drain with plenty of water.
2. Rinse both well plates thoroughly, first with tap water then with deionized water from your wash bottle.
3. Rinse all other pieces of equipment with deionized water and dry them.
4. Make sure all labeled 1-mL graduated pipets are returned to their appropriate test tubes.

Lab partner _____ Name _____

OBSERVATIONS: The marked boxes below correspond to the wells used in the experiment. Use the empty spaces for additional observations, if necessary.

PART A: $Fe^{3+}_{(aq)} + SCN^{-}_{(aq)} \leftrightarrows [FeNCS^{2+}]_{(aq)}$

A1 $Fe^{3+}_{(aq)}$ Color and intensity:			**D1** $SCN^{-}_{(aq)}$ Color and intensity:
	B2 Concentrated solution (at equilibrium) Color and intensity: *You might consider question 1 at this point.*	**C2 CONTROL (dilute solution at equil.)** Color and intensity:	
A3 Add $Fe^{3+}_{(aq)}$ Color and intensity before disturbance: Color and intensity after disturbance. Any ppt? Actual shift (→ or ←):	**B3 Add** $SCN^{-}_{(aq)}$ Color and intensity before disturbance: Color and intensity after disturbance. Any ppt? Actual shift (→ or ←):	**C3 Add** $AgNO_{3(aq)}$ Color and intensity before disturbance: Color and intensity after disturbance. Any ppt? Actual shift (→ or ←):	**D3 Add** $Fe(NO_3)_{3(s)}$ Color and intensity before disturbance: Color and intensity after disturbance. Any ppt? Actual shift (→ or ←):

Part B: **Pink Reactant ⇆ Blue Product**

Color when cooled: _____

Color at room temperature: _____

Color when warmed: _____